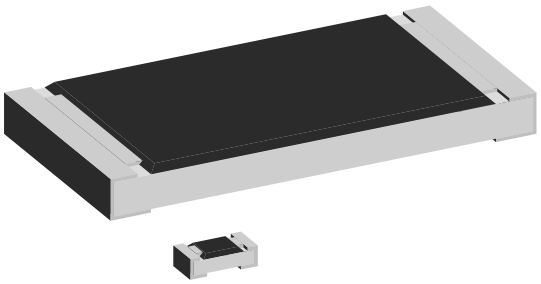
Standard Thick Film Chip Resistors



FEATURES

- Stability at different environmental conditions $\Delta R/R \leq 1$ % (1000 h rated power at 70 °C)
- 2 mm pitch packaging option for 0603 size
- AEC-Q200 qualified
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT

HALOGEN FREE

APPLICATIONS

- Automotive
- Industrial
- Telecommunication

LINKS TO ADDITIONAL RESOURCES

3D Models

D/CRCW e3 standard thick film chip resistors are the perfect choice for most fields of modern electronics where high reliability and stability are of major concern. Typical applications include automotive, telecommunications, and industrial.

TECHNICAL SPECIFICATIONS

DESCRIPTION	D10/ CRCW0402 e3	D11/ CRCW0603 e3	D12/ CRCW0805 e3	D25/ CRCW1206 e3	CRCW1210 e3	CRCW1218 e3	CRCW2010 e3	CRCW2512 e3
Imperial size	0402	0603	0805	1206	1210	1218	2010	2512
Metric size code	RR1005M	RR1608M	RR2012M	RR3216M	RR3225M	RR3246M	RR5025M	RR6332M
Resistance range	1 Ω to 10 MΩ; jumper (0 Ω)					1 Ω to 2.2 MΩ; jumper (0 Ω)	1 Ω to 10 MΩ; jumper (0 Ω)	
Resistance tolerance	± 5 %; ± 1 %							
Temperature coefficient	± 200 ppm/K; ± 100 ppm/K							
Rated dissipation, P_{70} (1)(2)	0.10 W	0.125 W	0.25 W	0.25 W	0.5 W	1.0 W	0.75 W	1.0 W
Operating voltage, $U_{max.}$ AC_{RMS}/DC (2)	75 V	75 V	150 V	200 V	200 V	200 V	400 V	500 V
Permissible film temperature, $\vartheta_{F\ max.}$ (1)	155 °C							
Operating temperature range	-55 °C to +155 °C							
Permissible voltage against ambient (insulation): 1 min, U_{ins}	100 V	100 V	200 V	300 V	300 V	300 V	300 V	300 V
Failure rate: $FIT_{observed}$	≤ 0.1 x 10^{-9}/h							

Notes

(1) Please refer to “Application Information” below

(2) Please refer to “Maximum Resistance Change at Rated Dissipation and Operating Voltage” table, see below

APPLICATION INFORMATION

When the resistor dissipates power, a temperature rise above the ambient temperature occurs, dependent on the thermal resistance of the assembled resistor together with the printed circuit board. The rated dissipation applies only if the permitted film temperature is not exceeded.

These resistors do not feature a limited lifetime when operated within the permissible limits. However, resistance value drift increasing over operating time may result in exceeding a limit acceptable to the specific application, thereby establishing a functional lifetime.

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MAXIMUM RESISTANCE CHANGE AT RATED DISSIPATION AND OPERATING VOLTAGE			
OPERATION MODE		**STANDARD**	**EXTENDED**
Rated dissipation, P_{70}	**D10/CRCW0402 e3**	0.063 W	0.10 W
	D11/CRCW0603 e3	0.10 W	0.125 W
	D12/CRCW0805 e3	0.125 W	0.25 W
Operating voltage, $U_{max.}$ AC_{RMS}/DC	**D10/CRCW0402 e3**	50 V	75 V
	D11/CRCW0603 e3	75 V	75 V
	D12/CRCW0805 e3	150 V	150 V
Resistance change		1 Ω to 10 MΩ	
Max. resistance change at P_{70}/$U_{max.}$ AC_{RMS}/DC for resistance change, $I\Delta R/RI$ after (1):	1000 h	≤ 1 %	≤ 2 %
	8000 h	≤ 2 %	-

Note

(1) Applies to components with stability class 1

TEMPERATURE COEFFICIENT AND RESISTANCE RANGE				
TYPE / SIZE	**TCR**	**TOLERANCE**	**RESISTANCE**	**E-SERIES**
D10/CRCW0402 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 200 ppm/K	± 1 %	1 Ω to 9.76 Ω	E24; E96
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.}$ = 1.5 A	≤ 20 mΩ	0 Ω	-
D11/CRCW0603 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 200 ppm/K	± 1 %	1 Ω to 9.76 Ω	E24; E96
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.}$ = 2.0 A	≤ 20 mΩ	0 Ω	-
D12/CRCW0805 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 200 ppm/K	± 1 %	1 Ω to 9.76 Ω	E24; E96
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.}$ = 2.5 A	≤ 20 mΩ	0 Ω	-
D25/CRCW1206 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 200 ppm/K	± 1 %	1 Ω to 9.76 Ω	E24; E96
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.}$ = 3.5 A	≤ 20 mΩ	0 Ω	-
CRCW1210 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 200 ppm/K	± 1 %	1 Ω to 9.76 Ω	E24; E96
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.}$ = 5.0 A	≤ 20 mΩ	0 Ω	-
CRCW1218 e3	± 200 ppm/K	± 5 %	1 Ω to 2.2 MΩ	E24
	± 200 ppm/K	± 1 %	1 Ω to 9.76 Ω	E24; E96
	± 100 ppm/K	± 1 %	1 Ω to 2.2 MΩ	E24; E96
	Jumper, $I_{max.}$ = 7.0 A	≤ 20 mΩ	0 Ω	-
CRCW2010 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 200 ppm/K	± 1 %	1 Ω to 9.76 Ω	E24; E96
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.}$ = 6.0 A	≤ 20 mΩ	0 Ω	-
CRCW2512 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 200 ppm/K	± 1 %	1 Ω to 9.76 Ω	E24; E96
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.}$ = 7.0 A	≤ 20 mΩ	0 Ω	-

Note

- The temperature coefficient of resistance (TCR) is not specified for 0 Ω jumpers

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PACKAGING

TYPE / SIZE	CODE	QUANTITY	PACKAGING STYLE	WIDTH	PITCH	PACKAGING DIMENSIONS
D10/CRCW0402 e3	ED = ET7	10 000	Paper tape acc. to IEC 60286-3, Type 1a	8 mm	2 mm	Ø 180 mm / 7"
	EE = EF4	50 000				Ø 330 mm / 13"
D11/CRCW0603 e3	EI = ET2	5000				Ø 180 mm / 7"
	ED = ET3	10 000				Ø 180 mm / 7"
	EE = ET8	50 000				Ø 330 mm / 13"
	EA = ET1	5000			4 mm	Ø 180 mm / 7"
	EC = ET6	20 000				Ø 330 mm / 13"
D12/CRCW0805 e3	EA = ET1	5000				Ø 180 mm / 7"
	EC = ET6	20 000				Ø 330 mm / 13"
D25/CRCW1206 e3	EA = ET1	5000				Ø 180 mm / 7"
	EC = ET6	20 000				Ø 330 mm / 13"
CRCW1210 e3	EA = ET1	5000				Ø 180 mm / 7"
	EC = ET6	20 000				Ø 330 mm / 13"
CRCW1218 e3	EK = ET9	4000	Blister tape acc. to IEC 60286-3, Type 2a	12 mm	4 mm	Ø 180 mm / 7"
CRCW2010 e3	EF = E02	4000				Ø 180 mm / 7"
CRCW2512 e3	EG = E67	2000			8 mm	Ø 180 mm / 7"
	EH = E82	4000			4 mm	

PART NUMBER AND PRODUCT DESCRIPTION

Part Number: CRCW0603562RFKEA
Part Number: CRCW06030000Z0EA

C	R	C	W	0	6	0	3	5	6	2	R	F	K	E	A

TYPE / SIZE	RESISTANCE	TOLERANCE	TCR	PACKAGING
CRCW0402 **CRCW0603** **CRCW0805** **CRCW1206** **CRCW1210** **CRCW1218** **CRCW2010** **CRCW2512**	**R** = decimal **K** = thousand **M** = million **0000** = jumper	**F** = ± 1 % **J** = ± 5 % **Z** = jumper	**K** = ± 100 ppm/K **N** = ± 200 ppm/K **0** = jumper	**EA, EC, ED, EE,** **EF, EG, EH, EI, EK** (E.. = lead (Pb)-free)

Product Description: D11/CRCW0603 100 562R 1 % ET1 e3
Product Description: D11/CRCW0603 0R0 ET1 e3

D11/CRCW0603	100	562R	1 %	ET1	e3
TYPE / SIZE	TCR	RESISTANCE	TOLERANCE	PACKAGING	LEAD (Pb)-FREE
D10/CRCW0402 **D11/CRCW0603** **D12/CRCW0805** **D25/CRCW1206** **CRCW1210** **CRCW1218** **CRCW2010** **CRCW2512**	± **100** ppm/K ± **200** ppm/K	**10R** = 10 Ω **562R** = 562 Ω **1M** = 1 MΩ **0R0** = jumper	± **1 %** ± **5 %**	**ET1, ET2, ET3,** **ET6, ET7, ET8,** **ET9, EF4, E02,** **E67, E82** (E.. = lead (Pb)-free)	**e3** = pure tin termination finish

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DESCRIPTION

Production is strictly controlled and follows an extensive set of instructions established for reproducibility. A cermet film layer and a glass-over are deposited on a high grade (Al_2O_3) ceramic substrate with its prepared inner contacts. A special laser is used to achieve the target value by smoothly fine trimming the resistive layer without damaging the ceramics. The resistor elements are covered by a protective coating designed for electrical, mechanical and climatic protection. The terminations receive a final pure matte tin on nickel plating.

The result of the determined production is verified by an extensive testing procedure on 100 % of the individual chip resistors. Only accepted products are laid directly into the tape in accordance with **IEC 60286-3 Type 1a and Type 2a** (1).

ASSEMBLY

The resistors are suitable for processing on automatic SMD assembly systems. They are suitable for automatic soldering using wave, reflow or vapor phase as shown in **IEC 61760-1** (1). The encapsulation is resistant to all cleaning solvents commonly used in the electronics industry, including alcohols, esters and aqueous solutions. The suitability of conformal coatings, potting compounds and their processes, if applied, shall be qualified by appropriate means to ensure the long-term stability of the whole system.

The resistors are RoHS-compliant, the pure matte tin plating provides compatibility with lead (Pb)-free and lead-containing soldering processes. Solderability is specified for 2 years after production or requalification. The permitted storage time is 20 years. The immunity of the plating against tin whisker growth has been proven under extensive testing.

MATERIALS

Vishay acknowledges the following systems for the regulation of hazardous substances:

- IEC 62474, Material Declaration for Products of and for the Electrotechnical Industry, with the list of declarable substances given therein (2)
- The Global Automotive Declarable Substance List (GADSL) (3)
- The REACH regulation (1907/2006/EC) and the related list of substances with very high concern (SVHC) (4) for its supply chain

The products do not contain any of the banned substances as per IEC 62474, GADSL, or the SVHC list, see www.vishay.com/how/leadfree.

Hence the products fully comply with the following directives:

- 2000/53/EC End-of-Life Vehicle Directive (ELV) and Annex II (ELV II)
- 2011/65/EU Restriction of the Use of Hazardous Substances Directive (RoHS) with amendment 2015/863/EU
- 2012/19/EU Waste Electrical and Electronic Equipment Directive (WEEE)

Vishay pursues the elimination of conflict minerals from its supply chain, see the Conflict Minerals Policy at www.vishay.com/doc?49037.

APPROVALS

The resistors are qualified according to AEC-Q200.

Where applicable, the resistors are tested in accordance with **EN 140401-802** which refers to **EN 60115-1**, **EN 60115-8** and the variety of environmental test procedures of the **IEC 60068** (1) series.

RELATED PRODUCTS

For more information about products with better TCR and tighter tolerance please refer to the “Lead (Pb)-Free Thick Film, Rectangular, Semi-Precision Chip Resistors” datasheet (www.vishay.com/doc?20036).

The D/CRCW with SnPb termination plating is designed for applications where lead bearing terminations are mandatory. For ordering D/CRCW with SnPb terminations please refer to latest edition of datasheet D/CRCW (www.vishay.com/doc?20008).

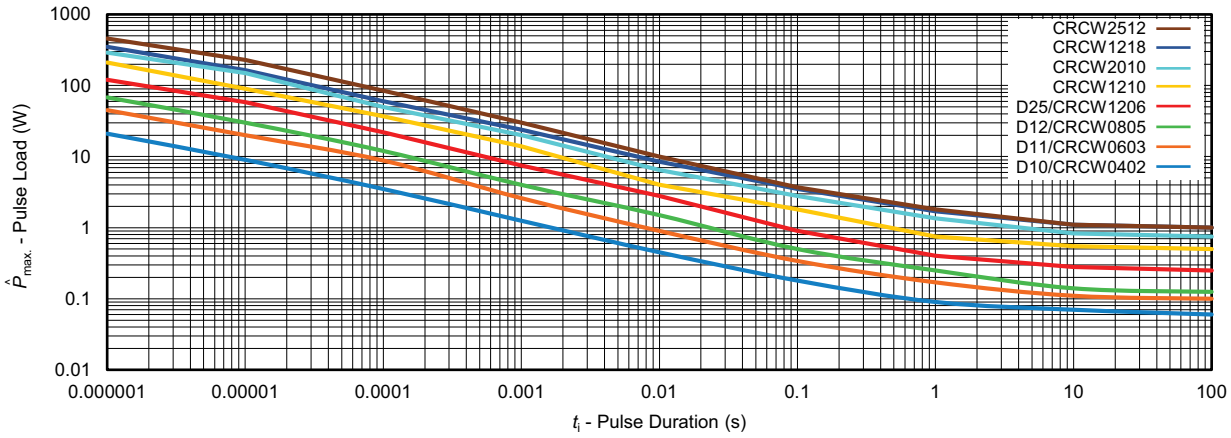
Notes

(1) The quoted IEC standards are also released as EN standards with the same number and identical contents
(2) The IEC 62474 list of declarable substances is maintained in a dedicated database, which is available at http://std.iec.ch/iec62474
(3) The Global Automotive Declarable Substance List (GADSL) is maintained by the American Chemistry Council and available at www.gadsl.org
(4) The SVHC list is maintained by the European Chemical Agency (ECHA) and available at http://echa.europa.eu/candidate-list-table

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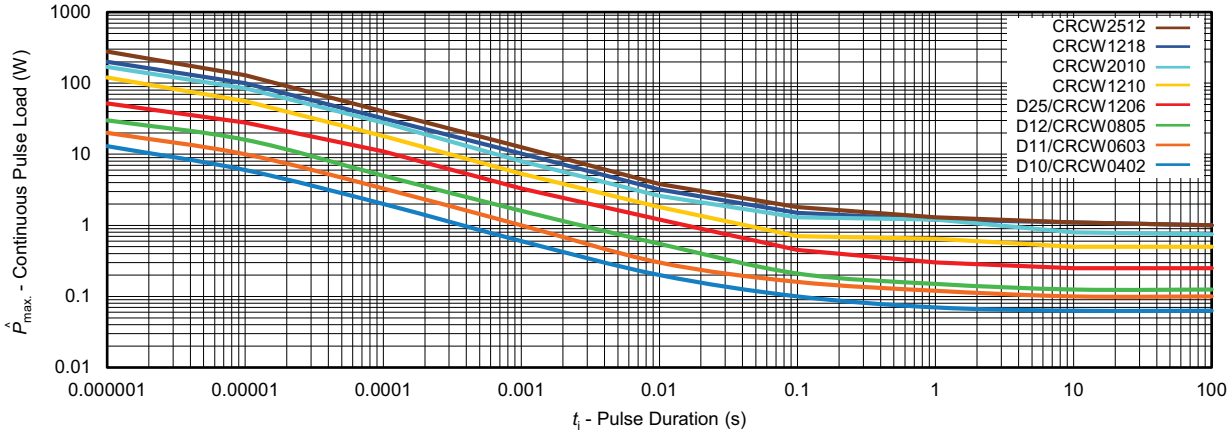
FUNCTIONAL PERFORMANCE

Single Pulse



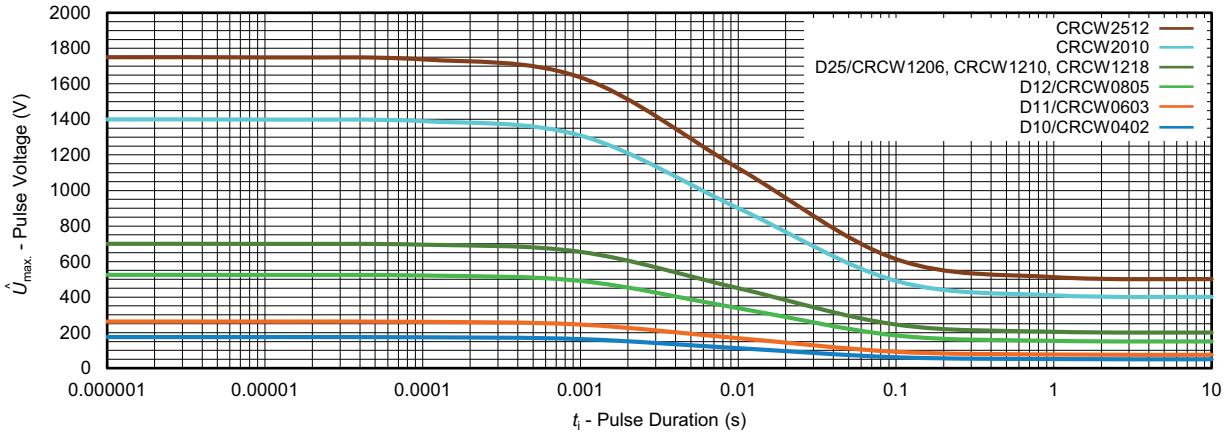
Maximum pulse load, single pulse; applicable if $\overline{P} \rightarrow 0$ and n < 1000 and $\hat{U} = \hat{U}_{max.}$; standard operation mode, for permissible resistance change equivalent to 8000 h operation

Continuous Pulse



Maximum pulse load, continuous pulses; applicable if $\overline{P} \leq P\,(\vartheta_{amb})$ and $\hat{U} = \hat{U}_{max.}$; standard operation mode, for permissible resistance change equivalent to 8000 h operation

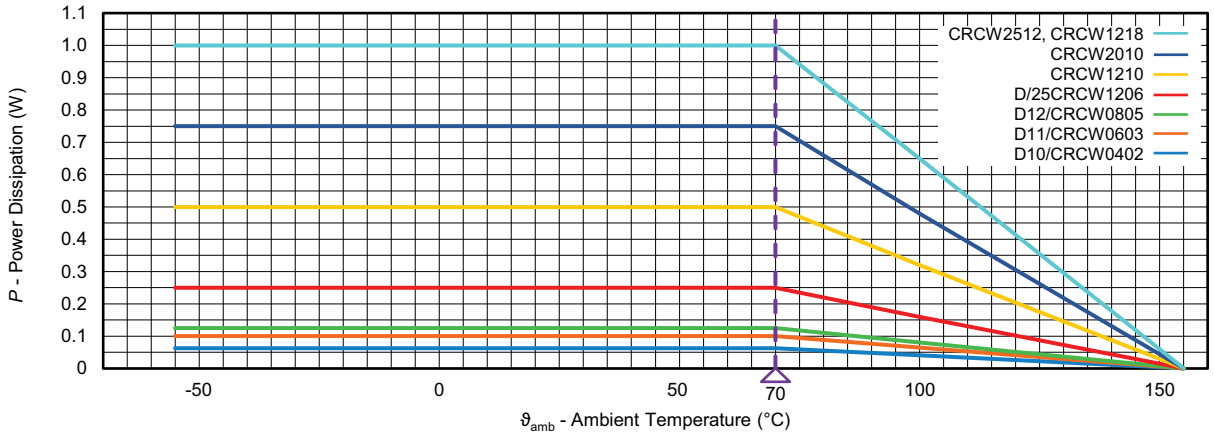
Pulse Voltage



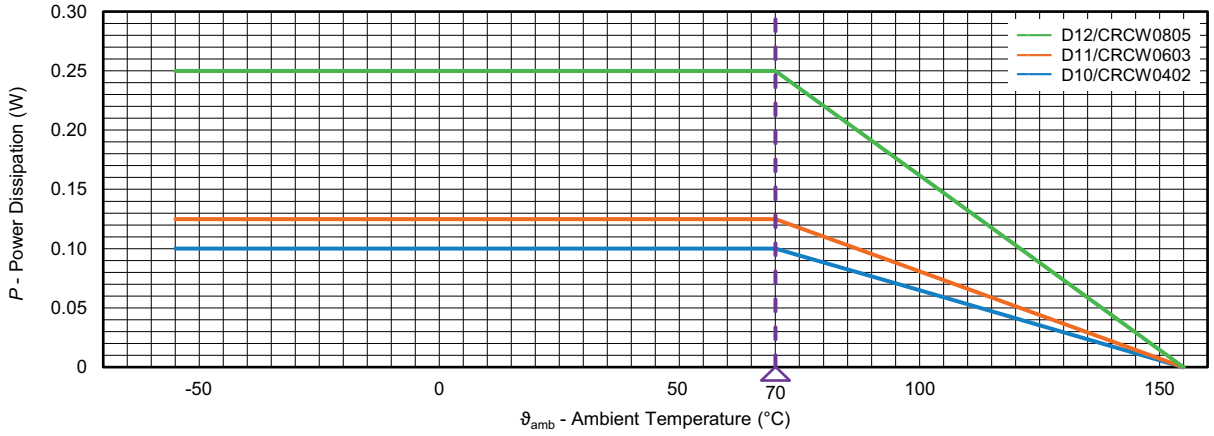
Maximum pulse voltage, single and continuous pulses; applicable if $\hat{P} = \hat{P}_{max.}$; standard operation mode, for permissible resistance change equivalent to 8000 h operation

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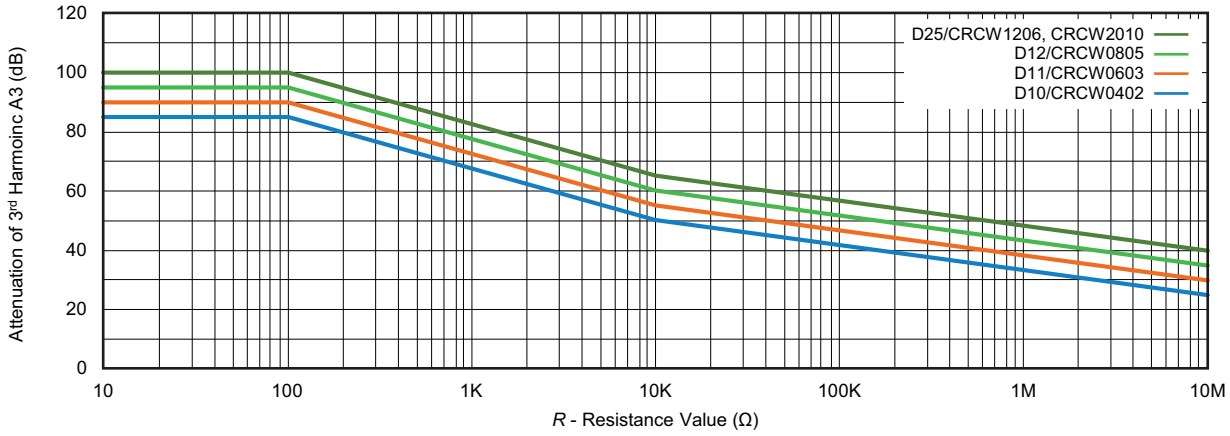
Derating



Derating - Extended Operation

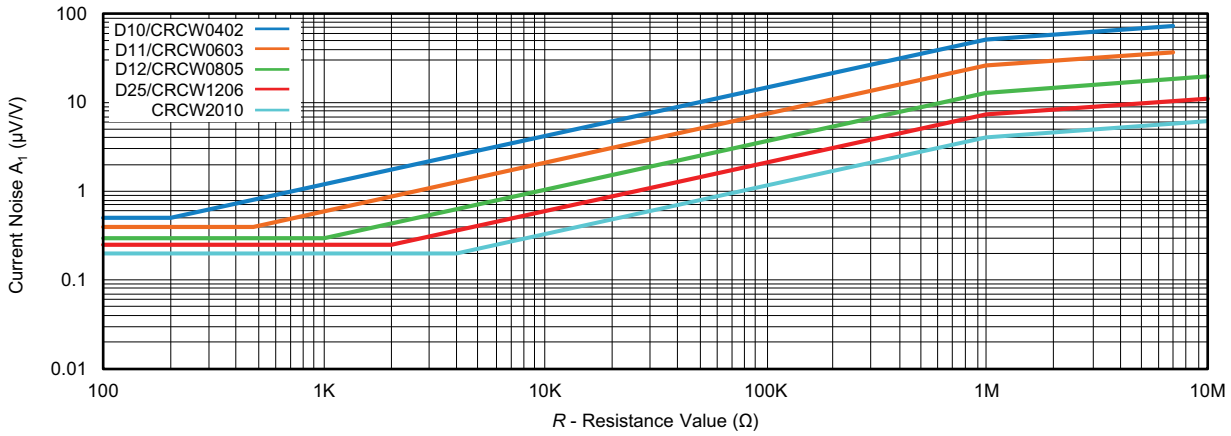


Non-Linearity



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Current Noise



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TESTS AND REQUIREMENTS

All executed tests are carried out in accordance with the following specifications:

EN 60115-1, generic specification

EN 60115-8, sectional specification

EN 140401-802, detail specification

IEC 60068-2-xx, test methods

The parameters stated in the Test Procedures and Requirements table are based on the required tests and permitted limits of EN 140401-802. The table presents only the most important tests, for the full test schedule refer to the documents listed above. However, some additional tests and a number of improvements against those minimum requirements have been included.

The testing also covers most of the requirements specified by EIA/IS-703 and JIS-C-5201-1.

The tests are carried out under standard atmospheric conditions in accordance with IEC 60068-1, 4.3, whereupon the following values are applied:

Temperature: 15 °C to 35 °C

Relative humidity: 25 % to 75 %

Air pressure: 86 kPa to 106 kPa (860 mbar to 1060 mbar).

A climatic category LCT / UCT / 56 is applied, defined by the lower category temperature (LCT), the upper category temperature (UCT), and the duration of exposure in the damp heat, steady state test (56 days). The components are mounted for testing on boards in accordance with EN 60115-8, 5.2.2 unless otherwise specified.

TEST PROCEDURES AND REQUIREMENTS

EN 60115-1 CLAUSE	IEC 60068-2 TEST METHOD	TEST	PROCEDURE	REQUIREMENTS PERMISSIBLE CHANGE (ΔR) [1]	
				STABILITY CLASS 1 OR BETTER	STABILITY CLASS 2 OR BETTER
			Stability for product types:	D10/CRCW0402 e3 to CRCW2512 e3	
			D/CRCW e3	1 Ω to 10 MΩ	
6.1	-	Measurements of resistance and tolerance	-	± 1 %	± 5 %
6.2	-	Temperature coefficient of resistance	At (20 / -55 / 20) °C and (20 / 155 / 20) °C	± 100 ppm/K; ± 200 ppm/K	± 200 ppm/K
7.1	-	Endurance at rated temperature 70 °C	$U = \sqrt{P_{70} \times R} \leq U_{max.}$; whichever is the less severe; 1.5 h on; 0.5 h off 70 °C; 1000 h 70 °C; 8000 h	± (1 % R + 0.05 Ω) ± (2 % R + 0.1 Ω)	± (2 % R + 0.1 Ω) ± (4 % R + 0.1 Ω)
		Endurance at rated temperature 70 °C Extended operation mode	$U = \sqrt{P_{70} \times R} \leq U_{max.}$; whichever is the less severe; 1.5 h on; 0.5 h off 70 °C; 1000 h	± (2 % R + 0.1 Ω)	
7.3	-	Endurance at maximum temperature	155 °C, 1000 h	± (1 % R + 0.05 Ω)	± (2 % R + 0.1 Ω)
10.4	78 (Cab)	Damp heat, steady state	(40 ± 2) °C; (93 ± 3) % RH; 56 days	± (1 % R + 0.05 Ω)	
10.5	67 (Cy)	Damp heat, steady state, accelerated Extended or standard operation mode depend on case size	(85 ± 2) °C; (85 ± 5) % RH; $U = \sqrt{0.1 \times P_{85} \times R}$ ≤ 100 V; 1000 h	± (1 % R + 0.05 Ω)	± (2 % R + 0.1 Ω)

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TEST PROCEDURES AND REQUIREMENTS

EN 60115-1 CLAUSE	IEC 60068-2 TEST METHOD	TEST	PROCEDURE	REQUIREMENTS PERMISSIBLE CHANGE (ΔR) [1]	
				STABILITY CLASS 1 OR BETTER	STABILITY CLASS 2 OR BETTER
			Stability for product types:	D10/CRCW0402 e3 to CRCW2512 e3	
			D/CRCW e3	1 Ω to 10 MΩ	
10.3	-	Climatic sequence:	-	± (1 % R + 0.05 Ω)	± (2 % R + 0.1 Ω)
10.3.4.2	2 (Bb)	Dry heat	125 °C; 16 h		
10.3.4.3	30 (Db)	Damp heat, cyclic	55 °C; ≥ 90 % RH; 24 h; 1 cycle		
10.3.4.4	1 (Ab)	Cold	-55 °C; 2 h		
10.3.4.5	13 (M)	Low air pressure	8.5 kPa; 2 h; (25 ± 10) °C; 55 °C; > 90 % RH		
10.3.4.6	30 (Db)	Damp heat, cyclic	5 days; 5 cycles		
10.3.4.7	-	DC load Standard operation mode	$U = \sqrt{P_{70} \times R} \leq U_{max.}$; 1 min		
-	1 (Ab)	Cold	-55 °C; 2 h	± (0.25 % R + 0.05 Ω)	± (0.5 % R + 0.05 Ω)
10.1	14 (Na)	Rapid change of temperature	30 min at -55 °C; and 30 min at 125 °C; 1000 cycles	± (1 % R + 0.05 Ω) no visible damage	
8.1	-	Short-term overload	$U = 2.5 \times \sqrt{P_{70} \times R} \leq 2 \times U_{max.}$; whichever is the less severe; 5 s	± (2 % R + 0.05 Ω)	
8.2	-	Single pulse high voltage overload, 10 µs/700 µs	Severity no. 4: $U = 10 \times \sqrt{P_{70} \times R} \leq 2 \times U_{max.}$; whichever is the less severe; 10 pulses 10 µs/700 µs	± (1 % R + 0.05 Ω) no visible damage	
		Single pulse high voltage overload, 10 µs/700 µs Extended operation mode	Severity no. 4: $U = 10 \times \sqrt{P_{70} \times R} \leq 2 \times U_{max.}$; whichever is the less severe; 10 pulses 10 µs/700 µs	± (2 % R + 0.1 Ω) no visible damage	
8.4	-	Periodic electric overload	$U = \sqrt{15 \times P_{70} \times R}$ or ≤ 2 x $U_{max.}$; whichever is the less severe; 0.1 s on; 2.5 s off; 1000 cycles	± (1 % R + 0.05 Ω) no visible damage	
		Periodic electric overload Extended operation mode	$U = \sqrt{15 \times P_{70} \times R}$ or ≤ 2 x $U_{max.}$; whichever is the less severe; 0.1 s on; 2.5 s off; 1000 cycles	± (2 % R + 0.1 Ω) no visible damage	
8.5	-	Electrostatic discharge (human body model)	IEC 61340-3-1 [1]; 3 pos. + 3 neg. discharges; ESD voltage acc. to the size: D10/CRCW0402 e3: 400 V D11/CRCW0603 e3: 800 V D12/CRCW0805 e3: 1000 V D25/CRCW1206 e3: 2000 V CRCW1210 e3 to CRCW2512 e3: ≥ 2000 V	± (1 % R + 0.05 Ω)	

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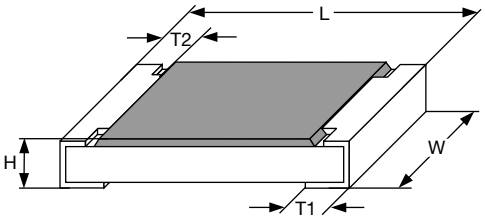
TEST PROCEDURES AND REQUIREMENTS					
EN 60115-1 CLAUSE	IEC 60068-2 TEST METHOD	TEST	PROCEDURE	REQUIREMENTS PERMISSIBLE CHANGE (ΔR) (1)	
				STABILITY CLASS 1 OR BETTER	STABILITY CLASS 2 OR BETTER
			Stability for product types:	D10/CRCW0402 e3 to CRCW2512 e3	
			D/CRCW e3	1 Ω to 10 MΩ	
9.11	6 (Fc)	Vibration	Endurance by sweeping; 10 Hz to 2000 Hz; no resonance; amplitude ≤ 1.5 mm or ≤ 200 m/s^2; 7.5 h	± (0.25 % R + 0.05 Ω) no visible damage	± (0.5 % R + 0.05 Ω) no visible damage
11.1	58 (Td)	Solderability	Solder bath method; Sn60Pb40 non-activated flux; (235 ± 5) °C (2 ± 0.2) s	Good tinning (≥ 95 % covered); no visible damage	
			Solder bath method; Sn96.5Ag3Cu0.5 non-activated flux; (245 ± 5) °C (3 ± 0.3) s		
11.2	58 (Td)	Resistance to soldering heat	Solder bath method (260 ± 5) °C; (10 ± 1) s	± (0.25 % R + 0.05 Ω)	± (0.5 % R + 0.05 Ω)
11.3	45 (XA)	Component solvent resistance	Isopropyl alcohol; 50 °C; method 2	No visible damage	
9.7	21 (Ue_3)	Shear (adhesion)	D10/CRCW0402 e3 and D11/CRCW0603 e3: 9 N D12/CRCW0805 e3 to CRCW2512 e3: 45 N	No visible damage	
9.8	21 (Ue_1)	Substrate bending	Depth 2 mm; 3 times	± (0.25 % R + 0.05 Ω) no visible damage, no open circuit in bent position	
12.2	-	Voltage proof	U = 1.4 x U_{ins}; 60 s	No flashover or breakdown	
12.4	-	Flammability, needle flame test	IEC 60695-11-5 (1); 10 s	No burning after 30 s	

Note

(1) The quoted IEC standards are also released as EN standards with the same number and identical contents

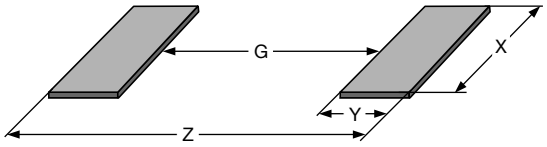
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DIMENSIONS



DIMENSIONS AND MASS						
TYPE / SIZE	L (mm)	W (mm)	H (mm)	T1 (mm)	T2 (mm)	MASS (mg)
D10/CRCW0402 e3	1.0 ± 0.05	0.5 ± 0.05	0.35 ± 0.05	0.25 ± 0.10	0.2 ± 0.10	0.65
D11/CRCW0603 e3	1.55 + 0.10 / - 0.05	0.85 ± 0.10	0.45 ± 0.05	0.3 ± 0.20	0.3 ± 0.20	2
D12/CRCW0805 e3	2.0 + 0.20 / - 0.10	1.25 ± 0.15	0.5 ± 0.10	0.3 + 0.20 / - 0.10	0.3 ± 0.20	5.5
D25/CRCW1206 e3	3.2 + 0.10 / - 0.20	1.6 ± 0.15	0.55 ± 0.05	0.45 ± 0.20	0.4 ± 0.20	10
CRCW1210 e3	3.2 ± 0.20	2.5 ± 0.20	0.55 ± 0.05	0.45 ± 0.20	0.4 ± 0.20	16
CRCW1218 e3	3.2 + 0.10 / - 0.20	4.6 ± 0.15	0.55 ± 0.05	0.45 ± 0.20	0.4 ± 0.20	29.5
CRCW2010 e3	5.0 ± 0.15	2.5 ± 0.15	0.6 ± 0.10	0.6 ± 0.20	0.6 ± 0.20	25.5
CRCW2512 e3	6.3 ± 0.20	3.15 ± 0.15	0.6 ± 0.10	0.6 ± 0.20	0.6 ± 0.20	40.5

SOLDER PAD DIMENSIONS



RECOMMENDED SOLDER PAD DIMENSIONS								
TYPE / SIZE	WAVE SOLDERING				REFLOW SOLDERING			
	G (mm)	Y (mm)	X (mm)	Z (mm)	G (mm)	Y (mm)	X (mm)	Z (mm)
D10/CRCW0402 e3	-	-	-	-	0.45	0.6	0.6	1.65
D11/CRCW0603 e3	0.65	1.10	1.25	2.85	0.75	0.75	1.00	2.25
D12/CRCW0805 e3	0.90	1.30	1.60	3.50	1.00	0.95	1.45	2.90
D25/CRCW1206 e3	1.40	1.40	1.95	4.20	1.50	1.05	1.80	3.60
CRCW1210 e3	1.80	1.45	2.95	4.70	1.70	1.10	2.80	3.90
CRCW1218 e3	1.80	1.30	5.10	4.40	1.90	1.10	4.90	4.10
CRCW2010 e3	3.40	1.65	2.85	6.70	3.50	1.45	2.80	6.40
CRCW2512 e3	4.60	1.60	3.65	7.80	4.75	1.45	3.50	7.65

Notes

- The given solder pad dimensions reflect the considerations for board design and assembly as outlined e.g in standards IEC 61188-5-x [(1)] or in publication IPC-7351.
  Still, the given solder pad dimensions will be found adequate for most general applications

(1) The quoted IEC standards are also released as EN standards with the same number and identical contents

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REVISION HISTORY	
REVISION DATE	**DESCRIPTION**
14-Apr-2026	Page 2, “Temperature Coefficient And Resistance Range” table: - Added TCR ± 200 ppm/K to 1R-9R76, ± 1 % components Page 3, “Packaging” table: - Change EG = E67 code quantity to 2000 pcs Page 8, “Test Procedures And Requirements” table: - Added TCR ± 200 ppm/K to temperature coefficient of resistance for stability class 1 or better components Page 10: - Introduction of “Revision History” table

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